Figure 10-6 shows the simplest form of association. A solid line simply declares that two entities are related somehow. It is possible to add more information, however.

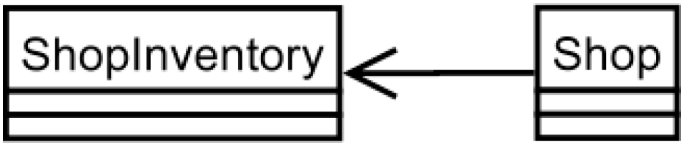


Figure 10-7. *Shop references Inventory*

By adding an open arrow, we've added a direction to the association; in this case, we're stating that Shop will have a reference to ShopInventory, but ShopInventory does not have a reference to Shop. This is *unidirectional*. If both classes held references to each other, then we would have arrowheads on both sides, showing a bi-directional relationship.

Of course, a class property might contain an array of references to another (to represent a collection), and it is possible to quantify associations by putting numbers (or an asterisk for any number) next to each class in the diagram. ShopInventory probably has more than one book in it, and we could show the association between ShopInventory and Book like this:

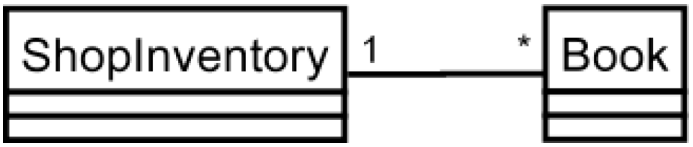


Figure 10-8. *One ShopInventory, many Books*

The figure above shows that our model now contains a single ShopInventory instance that refers to zero or more Book instances. It is also possible to show ranges in this form, if ShopInventory has a maximum capacity of 1000 books, we could replace the asterisk next to Book with 0..1000.